Robust Benchmark-Relative Tests of Weak-Form Efficiency in Latin American Equity ETFs

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Abstract

This paper evaluates weak-form efficiency in four Latin American equity ETFs (EWZ, EWW, ECH, GXG) using benchmark-relative out-of-sample forecasting tests. The estimand is forecast-loss difference versus a random-walk baseline under a common lagged information set. We implement expanding-window evaluation with horizon gap controls, leakage assertions, and multiple loss metrics (RMSE, MAE, directional accuracy) across six (L, H) configurations. Benchmarks include random walk, historical mean, linear autoregressive, ridge, MLP, and random-forest models, with conditional diffusion evaluated under the same target definition and window protocol.

Random walk remains the lowest-RMSE model in all tested configurations. At short horizon ($H = 1$), some simple alternatives are close to random walk but gains are weak or unstable across settings. Diffusion underperforms benchmark models materially and is significantly worse than random walk in all Diebold-Mariano comparisons in this implementation. Diffusion seed and feature ablations indicate substantial sensitivity, and XAI reliance patterns are not stable enough for strong structural interpretation. Findings are therefore interpreted as disciplined evidence of limited extractable short-horizon predictability, not as proof of exploitable inefficiency.

1 Introduction

Weak-form market efficiency remains a core benchmark in empirical finance. In emerging markets, however, integration frictions, episodic illiquidity, and structural regime changes motivate repeated reassessment of short-horizon predictability (**????**).

This study asks one focused question:

> Do diffusion-based forecasts deliver robust out-of-sample loss reductions relative to strong benchmark models for Latin American ETF cumulative returns?

The empirical contribution is a benchmark-disciplined comparison across EWZ, EWW, ECH, and GXG over six (L, H) configurations with expanding-window evaluation and Diebold-Mariano inference against random walk. The methodological contribution is a leakage-audited, horizon-consistent workflow that combines classical baselines, nonlinear baselines, conditional diffusion, and seed/feature sensitivity analysis under a common target definition. The interpretability contribution is a stability-oriented XAI layer that treats feature reliance as predictive model dependence, not causal mechanism.

Inference is intentionally conservative. The design supports statements about forecast association and incremental predictive content; it does not identify causal effects, structural transmission parameters, or implementable profitability after transaction costs.

2 Literature Positioning and Gap

2.1 Relevant Literature Strands

The manuscript is positioned at the intersection of five strands.

1. **Weak-form efficiency and return predictability tests**: foundational evidence on random-walk violations and instability of forecasting gains (**????**).
2. **Emerging-market efficiency and integration**: evidence that return behavior varies with integration and local information regimes (**??**).
3. **Latin American efficiency evidence**: mostly based on variance-ratio, complexity, or adaptive-market test batteries rather than benchmark-first forecast competitions (**????**).
4. **Machine-learning return modeling**: nonlinear models can improve fit in some settings but require strong out-of-sample discipline (**?**).
5. **Diffusion-model methodology**: denoising diffusion and score-based methods provide flexible distributional modeling, and early finance applications are emerging, but weak-form efficiency implications under explicit benchmark tests remain open (**???**).

2.2 Gap Taxonomy

The operational gap is sharpened along five dimensions.

- **Theoretical gap**: weak-form discussion is often framed as a binary rejection/acceptance exercise, not as relative forecast-loss performance under identical information sets.
- **Empirical gap**: Latin American studies are concentrated in index-level dependence testing; ETF-level multi-model out-of-sample evidence is limited in this implementation domain.
- **Methodological gap**: many studies emphasize standalone test batteries instead of unified forecast competitions with equal feature sets, identical splits, and explicit benchmark inference.
- **Data/context gap**: this project uses tradable regional ETF proxies with harmonized preprocessing and horizon-specific cumulative-return targets.
- **Identification gap**: descriptive dependence, predictive association, and economically exploitable inefficiency are frequently conflated.

2.3 Defensible Gap Statement

This paper addresses a narrow and testable gap: whether a conditional diffusion forecaster can deliver *incremental out-of-sample predictive accuracy* over a random-walk benchmark for Latin American ETF returns under horizon-consistent holdout design and explicit forecast-accuracy inference. The contribution is therefore comparative and inferential, not structural or causal.

3 Data and Variable Construction

3.1 Data Source and Scope

Raw adjusted-close prices and volumes are downloaded from Yahoo Finance through `yfinance`. The empirical assets are EWZ (Brazil), EWW (Mexico), ECH (Chile), and GXG (Colombia). Frequency is daily.

Table 1: Data overview.

Item	Value
Assets	4
Observations (daily)	2264
Start date	2015-01-02
End date	2023-12-29
Average active mask	1.000

Raw data source: Yahoo Finance via yfinance; assets: EWZ, EWW, ECH, GXG.

Table 1 reports the effective sample (2015-01-02 to 2023-12-29, 2,264 daily observations). The preprocessing pipeline is deterministic: prices are forward-filled, returns are computed from adjacent adjusted closes, and a tradability mask suppresses inactive observations (missing price or non-positive volume).

3.2 Variable Construction

For asset i at date t, let $P_{i,t}$ denote adjusted close and $V_{i,t}$ traded volume. The tradability mask is

$$m_{i,t} = \mathbb{1}(P_{i,t} \text{ observed and } V_{i,t} > 0)\,. \tag{1}$$

The daily log return is

$$r_{i,t} = \log\left(\frac{P_{i,t}}{P_{i,t-1}}\right) m_{i,t}. \tag{2}$$

Following **?**, the illiquidity proxy is

$$\text{illiq}_{i,t} = \frac{|r_{i,t}|}{P_{i,t}V_{i,t} + \varepsilon}, \quad \varepsilon = 10^{-8}, \tag{3}$$

and standardized with a 60-day rolling window:

$$z_{i,t}^{\text{illiq}} = \frac{\text{illiq}_{i,t} - \mu_{i,t}^{(60)}}{\sigma_{i,t}^{(60)} + \varepsilon}. \tag{4}$$

with $\mu_{i,t}^{(60)}$ and $\sigma_{i,t}^{(60)}$ denoting rolling mean and standard deviation. Values undefined in early windows are set to zero in the processed feature matrix.

The model context at forecast origin t is

$$\mathcal{X}_t = \left\{ r_{i,\tau}, m_{i,\tau}, z_{i,\tau}^{\text{illiq}} \right\}_{i=1,\ldots,N;\, \tau=t-L,\ldots,t-1}. \tag{5}$$

The H-day forecast target is cumulative return

$$y_{i,t}^{(H)} = \sum_{j=0}^{H-1} r_{i,t+j}. \tag{6}$$

which matches the implementation in the dataset loader used by all competing models.

4 Econometric Strategy and Identification

4.1 Object of Inference

The core estimand is benchmark-relative out-of-sample loss improvement versus random walk. For candidate model m,

$$\Delta\mathcal{L}_m = \mathbb{E}[\ell(e_t^{\text{rw}}) - \ell(e_t^m)]\,, \tag{7}$$

where $\ell(\cdot)$ is forecast loss and $e_t^m = y_t - \hat{y}_t^m$. Positive values indicate lower loss than random walk. The design identifies predictive association only.

4.2 Temporal Integrity and Leakage Controls

Leakage safeguards are imposed at preprocessing, sample construction, and validation stages.

1. Processed feature indices are required to be time-sorted, unique, and aligned across returns, mask, and illiquidity inputs.

2. The illiquidity standardization uses strictly lagged rolling moments in code, so statistics at t are computed from data up to $t-1$.

3. For each origin i, context uses $[i-L, \ldots, i-1]$ and target uses $[i, \ldots, i+H-1]$.

4. Expanding-window evaluation inserts a horizon gap $H-1$ between training and validation origins, and assertions fail if target windows overlap.

4.3 Validation Design

Instead of a single split, each (L, H) configuration is evaluated on multiple expanding windows. Let $w \in \{1, \ldots, W\}$ index windows. For each w, models are fitted on a chronological training

block and evaluated on a later validation block. Per-window losses are stored and aggregated, and W is reported with each summary metric.

4.4 Competing Forecast Models

All models forecast the same cumulative-return target from comparable lagged information sets:

1. **Random walk** ($\hat{y}_{i,t}^{(H)} = 0$).
2. **Historical mean** (mean lagged return in lookback window).
3. **Linear autoregressive baseline** (asset-wise linear AR on lagged returns).
4. **Ridge regression** on flattened context with time-aware inner validation for α.
5. **Shallow MLP baseline** with train-only scaling and early stopping.
6. **Random-forest baseline** (parsimonious tree depth and leaf-size controls).
7. **Conditional diffusion model**.

For diffusion with schedule $\{\beta_s\}_{s=1}^{S}$, define $\alpha_s = 1 - \beta_s$, $\bar{\alpha}_s = \prod_{u=1}^{s} \alpha_u$, and

$$y_s = \sqrt{\bar{\alpha}_s}\, y_0 + \sqrt{1 - \bar{\alpha}_s}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{8}$$

The denoiser ϵ_θ is trained by

$$\min_{\theta} \; \mathbb{E}_{y_0,\epsilon,s}\left[\|\epsilon - \epsilon_\theta(y_s, s, \mathcal{X}_t)\|_2^2\right]. \tag{9}$$

Training uses fixed conservative hyperparameters, seed control, and an inner time-ordered validation split with early stopping.

4.5 Loss Metrics and Comparative Inference

Point-forecast losses include RMSE and MAE:

$$\mathrm{RMSE} = \sqrt{\frac{1}{TN}\sum_{t=1}^{T}\sum_{i=1}^{N}\left(y_{i,t}^{(H)} - \hat{y}_{i,t}^{(H)}\right)^2}, \quad \mathrm{MAE} = \frac{1}{TN}\sum_{t=1}^{T}\sum_{i=1}^{N}\left|y_{i,t}^{(H)} - \hat{y}_{i,t}^{(H)}\right|. \tag{10}$$

Directional accuracy is reported as a complementary diagnostic.

Forecast comparison uses Diebold–Mariano tests (**?**) with

$$d_t = \ell(e_t^{\mathrm{rw}}) - \ell(e_t^{m}), \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{v}_d/T}}. \tag{11}$$

Under this sign convention, negative DM indicates higher loss for the candidate model than random walk.

4.6 Robustness and Sensitivity Design

Robustness dimensions include horizon (H), lookback (L), window-to-window stability, diffusion seed sensitivity, and feature-set ablations (`full`, `no_illiquidity`, `returns_only`). XAI analysis is framed as model reliance and explanation stability, not causality.

5 Empirical Results

5.1 Descriptive Properties

Table 2: Summary statistics of daily log returns.

ETF	Mean	Std. Dev.	Skewness	Kurtosis	ACF(1)
ECH	-0.00002	0.01733	-0.624	10.600	-0.012
EWW	0.00016	0.01597	-1.068	9.448	-0.010
EWZ	0.00018	0.02365	-1.137	13.272	-0.133
GXG	-0.00020	0.01725	-1.214	16.423	0.146

Table 3: Stationarity and serial dependence diagnostics.

ETF	ADF p (price)	ADF p (return)	LB p (return)	LB p (return^2)
ECH	0.348	0.000	0.000	0.000
EWW	0.805	0.000	0.000	0.000
EWZ	0.094	0.000	0.000	0.000
GXG	0.050	0.000	0.000	0.000

ADF is Augmented Dickey-Fuller test; LB is Ljung-Box at lag 10.

Table 4: Pairwise correlation matrix of daily log returns.

ETF	ECH	EWW	EWZ	GXG
ECH	1.000	0.586	0.568	0.581
EWW	0.586	1.000	0.662	0.611
EWZ	0.568	0.662	1.000	0.578
GXG	0.581	0.611	0.578	1.000

Table 2 shows near-zero unconditional mean returns with heavy tails. Table 3 confirms stationary returns, mostly non-stationary price levels, and strong serial dependence in squared returns. Table 4 indicates moderate cross-market comovement (off-diagonal correlations approximately 0.57–0.66), consistent with regional integration without near-collinearity.

5.2 Baseline Benchmark Performance

Random walk is the lowest-RMSE benchmark in all six (L, H) configurations (Tables 5 and 6 and fig. 1). At $H = 1$, random-forest and linear-AR baselines are close to random walk in level terms but do not dominate consistently across lookback settings. At $H = 5$, historical mean

Table 5: Out-of-sample RMSE by lookback window and forecast horizon.

L	H	AR Linear	Diffusion	Historical Mean	MLP	Random Forest	Random Walk	Ridge
10	1	0.02016	0.48328	0.02093	0.09460	0.02011	0.02003	0.02026
10	5	0.04630	0.64470	0.04629	0.11469	0.04657	0.04623	0.04659
21	1	0.02020	0.48483	0.02055	0.16236	0.02019	0.02006	0.02051
21	5	0.04633	0.47032	0.04650	0.20417	0.04664	0.04626	0.04737
42	1	0.02027	0.55543	0.02035	0.35766	0.02023	0.02013	0.02090
42	5	0.04653	0.55365	0.04652	0.34519	0.04689	0.04646	0.04872

Lower values indicate better forecast accuracy.

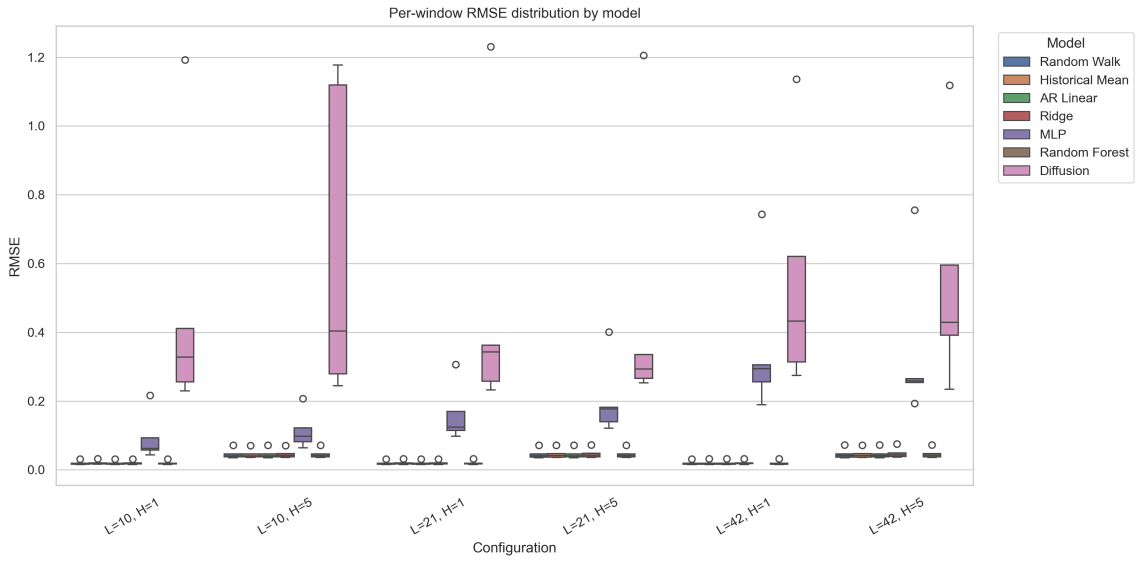


Figure 1: Per-window RMSE distribution by model and configuration.

Table 6: Average losses and directional accuracy across expanding windows.

L	H	Model	RMSE	MAE	Dir. Acc.	Windows
10	1	Random Walk	0.02003	0.01436	0.007	5
10	1	Random Forest	0.02011	0.01436	0.513	5
10	1	AR Linear	0.02016	0.01448	0.511	5
10	1	Ridge	0.02026	0.01459	0.499	5
10	1	Historical Mean	0.02093	0.01518	0.503	5
10	1	MLP	0.09460	0.06558	0.489	5
10	1	Diffusion	0.48328	0.38319	0.501	5
10	5	Random Walk	0.04623	0.03437	0.000	5
10	5	Historical Mean	0.04629	0.03444	0.505	5
10	5	AR Linear	0.04630	0.03441	0.506	5
10	5	Random Forest	0.04657	0.03453	0.509	5
10	5	Ridge	0.04659	0.03463	0.517	5
10	5	MLP	0.11469	0.08301	0.502	5
10	5	Diffusion	0.64470	0.51013	0.505	5
21	1	Random Walk	0.02006	0.01440	0.007	5
21	1	Random Forest	0.02019	0.01441	0.512	5
21	1	AR Linear	0.02020	0.01452	0.511	5
21	1	Ridge	0.02051	0.01480	0.492	5
21	1	Historical Mean	0.02055	0.01484	0.508	5
21	1	MLP	0.16236	0.11591	0.488	5
21	1	Diffusion	0.48483	0.38419	0.491	5
21	5	Random Walk	0.04626	0.03446	0.000	5
21	5	AR Linear	0.04633	0.03451	0.508	5
21	5	Historical Mean	0.04650	0.03460	0.511	5
21	5	Random Forest	0.04664	0.03476	0.476	5
21	5	Ridge	0.04737	0.03531	0.498	5
21	5	MLP	0.20417	0.14775	0.498	5
21	5	Diffusion	0.47032	0.37295	0.504	5
42	1	Random Walk	0.02013	0.01446	0.007	5
42	1	Random Forest	0.02023	0.01449	0.498	5
42	1	AR Linear	0.02027	0.01459	0.509	5
42	1	Historical Mean	0.02035	0.01466	0.503	5
42	1	Ridge	0.02090	0.01513	0.505	5
42	1	MLP	0.35766	0.26714	0.505	5
42	1	Diffusion	0.55543	0.44136	0.493	5
42	5	Random Walk	0.04646	0.03465	0.000	5
42	5	Historical Mean	0.04652	0.03468	0.503	5
42	5	AR Linear	0.04653	0.03469	0.507	5
42	5	Random Forest	0.04689	0.03509	0.479	5
42	5	Ridge	0.04872	0.03652	0.492	5
42	5	MLP	0.34519	0.25755	0.498	5
42	5	Diffusion	0.55365	0.43909	0.499	5

and AR-linear models are again close to random walk, while more flexible ML models (MLP) degrade materially.

5.3 Diffusion Model Performance

Diffusion forecasts are substantially worse than benchmark models in this implementation. Average RMSE for diffusion is approximately 0.47–0.64 across configurations, versus approximately 0.02 for $H = 1$ random-walk benchmarks and approximately 0.046 for $H = 5$ random-walk benchmarks (Table 5).

5.4 Formal Comparative Inference

Table 7: Diebold-Mariano comparison versus random walk.

Model	Mean DM statistic	Significant at 5%
AR Linear	-1.297	2 / 6
Diffusion	-21.694	6 / 6
Historical Mean	-1.594	3 / 6
MLP	-11.505	6 / 6
Random Forest	-2.002	3 / 6
Ridge	-4.704	5 / 6

Negative DM means the candidate model has higher loss than random walk.

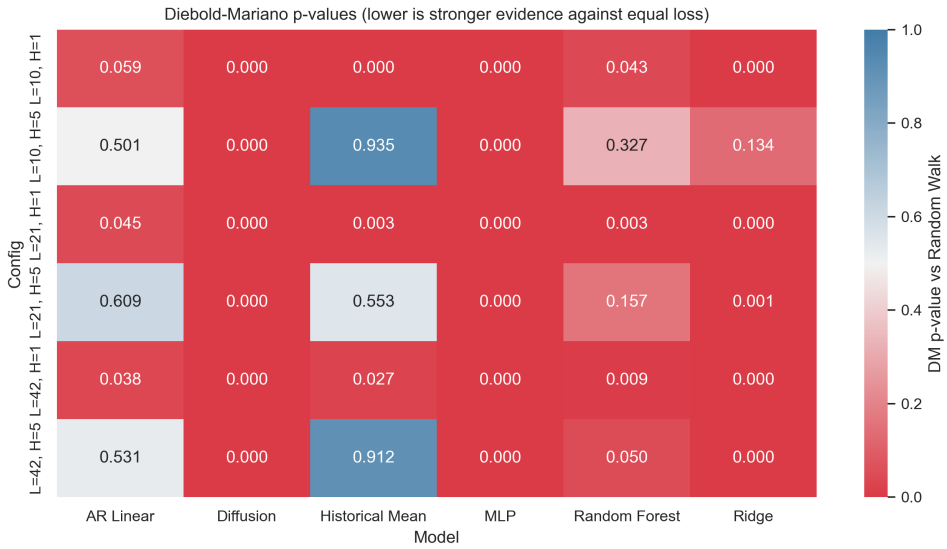
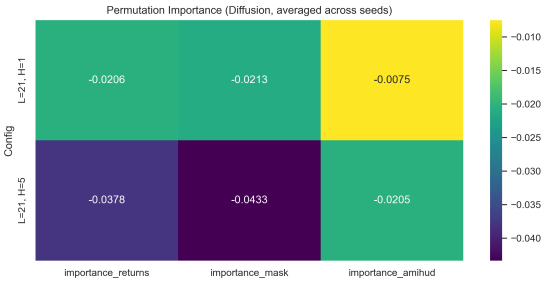
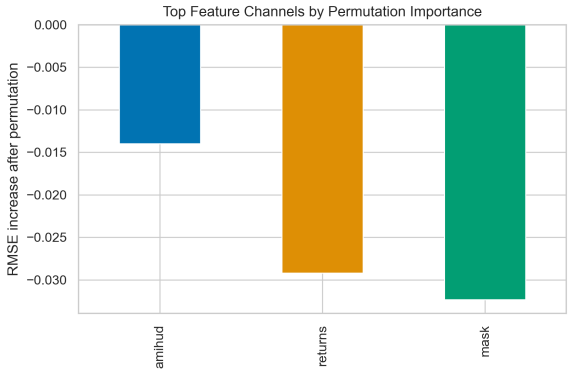


Figure 2: Diebold–Mariano p-values versus random walk.

Table 7 and fig. 2 confirms that diffusion is significantly worse than random walk in all six configurations under DM tests. Some baseline models are statistically indistinguishable from random walk at $H = 5$, while ridge and MLP are typically inferior.



(a) Average permutation importance



(b) Top feature channels

Figure 3: Diffusion model-reliance diagnostics from seed-level permutation analysis.

Table 8: Permutation-importance max z-score sensitivity across diffusion seeds.

L	H	Seed	Max z-score	Above 2.5 threshold
21	1	0	0.777	No
21	1	1	1.148	No
21	1	2	1.032	No
21	5	0	1.224	No
21	5	1	0.990	No
21	5	2	1.018	No

Table 9: XAI stability across seed pairs (Spearman correlation).

L	H	Seed pair	Spearman rho
21	1	0-1	0.500
21	1	0-2	0.500
21	1	1-2	-0.500
21	5	0-1	-0.500
21	5	0-2	1.000
21	5	1-2	-0.500

5.5 XAI Findings

XAI outputs are interpreted as model reliance only. Table 8 shows no seed with max-z above 2.5, and Table 9 shows unstable rank correlation across some seed pairs. Therefore, explanations are treated as fragile and non-structural.

5.6 Robustness Checks and Sensitivity

Table 10: Evaluation and leakage-control summary.

Item	Value
Evaluation windows	5
Total model-window evaluations	210
Leakage checks passed	True
Distinct benchmark models	6

Table 11: Diffusion sensitivity to feature-set ablations across seeds.

L	H	Feature set	RMSE	MAE	RMSE std.
21	1	full	0.79927	0.63577	0.02941
21	1	returns_only	0.82520	0.65499	0.07647
21	1	no_illiquidity	1.06475	0.84563	0.18551
21	5	returns_only	0.85052	0.67401	0.26983
21	5	no_illiquidity	0.98617	0.78908	0.36491
21	5	full	1.02785	0.82249	0.22140

Table 12: Diffusion seed-level sensitivity by horizon and feature set.

L	H	Feature set	Seed	RMSE	MAE
21	1	full	0	0.83312	0.65821
21	1	full	1	0.77999	0.61559
21	1	full	2	0.78470	0.63352
21	1	no_illiquidity	0	1.17537	0.93204
21	1	no_illiquidity	1	0.85059	0.67382
21	1	no_illiquidity	2	1.16830	0.93103
21	1	returns_only	0	0.80995	0.64342
21	1	returns_only	1	0.75751	0.60127
21	1	returns_only	2	0.90815	0.72029
21	5	full	0	1.15033	0.91929
21	5	full	1	0.77227	0.61856
21	5	full	2	1.16094	0.92963
21	5	no_illiquidity	0	1.18293	0.94173
21	5	no_illiquidity	1	0.56511	0.45356
21	5	no_illiquidity	2	1.21047	0.97196
21	5	returns_only	0	1.14319	0.90423
21	5	returns_only	1	0.79671	0.63215
21	5	returns_only	2	0.61164	0.48564

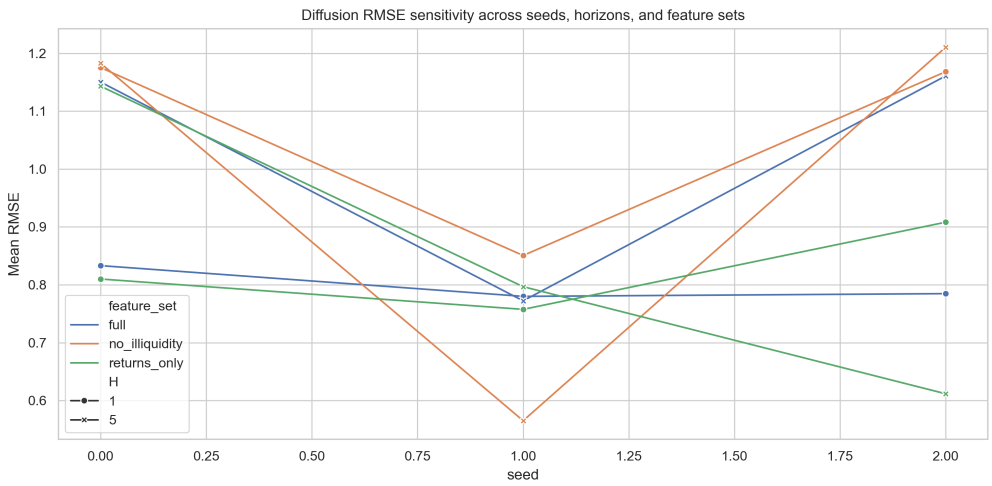


Figure 4: Diffusion sensitivity across seeds, horizons, and feature sets.

Leakage assertions pass for all evaluation windows (Table 10). Diffusion ablations in Tables 11 and 12 and fig. 4 indicate meaningful sensitivity to feature design and seed choice, with no robust improvement over benchmark models.

5.7 EMH Interpretation

Under a benchmark-relative predictability interpretation of weak-form EMH, the present evidence is consistent with limited extractable out-of-sample predictive gains beyond simple benchmarks in this dataset and design. This is a predictive, not structural, statement: the analysis does not establish exploitable abnormal profits, causal inefficiency mechanisms, or post-cost economic significance.

6 Limitations and Remaining Robustness Boundaries

The revised pipeline materially improves temporal integrity, benchmark breadth, and window-level evaluation; however, several boundaries remain.

1. **Economic significance is not tested**: no transaction-cost-adjusted trading strategy is implemented, so forecast comparisons should not be interpreted as profitability evidence.

2. **Model class coverage remains incomplete**: ARIMA/ARIMAX, GARCH-family, and VAR/VECM benchmarks are not yet included in the production loop.

3. **Hyperparameter search is conservative**: diffusion and MLP settings use leakage-safe but limited tuning; nested time-series hyperparameter selection is not yet implemented.

4. **Distributional validation is limited**: the current analysis emphasizes point-loss metrics; calibration-focused scoring and distributional diagnostics remain future work.

5. **Inference depth**: DM comparisons are reported, but Model Confidence Set procedures (**?**) are not yet integrated.

6. **Data scope**: conclusions are specific to four ETF proxies and the 2015–2023 sample, and may not generalize to other assets, frequencies, or post-2023 regimes.

7 Conclusion

This paper evaluates weak-form EMH through benchmark-relative forecast testing, not structural market-efficiency identification. Under expanding-window evaluation, leakage controls, multiple loss metrics, and a stronger benchmark family, random walk remains the best-performing model by RMSE across all tested (L, H) settings.

Conditional diffusion does not provide incremental predictive gains in this implementation and is significantly worse than random walk in all Diebold–Mariano comparisons. Diffusion ablation and seed analyses also indicate sensitivity rather than stable superiority. XAI outputs, interpreted as model reliance diagnostics, are not stable enough to support strong economic narratives.

The main contribution is therefore methodological: a transparent, reproducible, benchmark-disciplined pipeline for weak-form predictability testing in Latin American ETF proxies, including explicit robustness boundaries and interpretation discipline. Stronger claims would require broader benchmark classes, deeper hyperparameter validation, and an explicit post-cost economic significance layer.

A Supplementary Diagnostics and Reproducibility Notes

The robust experiment loop writes diagnostics and evaluation outputs to `reports/tables/`, including:

1. `per_window_losses.csv` (window-level RMSE, MAE, directional accuracy);
2. `leakage_checks.csv` (horizon-overlap assertions);
3. `diffusion_training_stability.csv` (best-epoch and validation-loss summaries);
4. `diffusion_seed_sensitivity.csv` and `diffusion_ablation.csv`;
5. `xai_seed_sensitivity.csv` and `xai_stability.csv`.

Manuscript asset generation is scripted via `scripts/build_manuscript_assets.py`, which converts report CSV files into `tables/generated/*.tex` and copies standardized figures to `figures/generated/`. Overleaf package export is automated by `scripts/build_working_paper.py`.